



OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	Two-Lane Highway Steel-Concrete Composite Bridge
Project Location	Nagpur, Maharashtra
Author / Designer	Structural Engineer
Reviewer	Chief Design Engineer
Organization	Indian Institute of Technology Bombay
Client Name and Organization	Ministry of Road Transport & Highways
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Executive Summary

This section provides a concise executive overview of the bridge geometry, design inputs, governing structural loads, and verified design outcomes.

Project Overview

Table 0.1: Project Overview and Structural Summary

Parameter / Metric	Design Specification / Result
Bridge Type	Highway Bridge
Design Standards	IRC 5, IRC 6, IRC 22, IRC 24, IRC 112, IS 800
Span	24.0 m
Carriageway Width	7.5 m
Number of Girders	4
Girder Spacing (c/c)	2.1 m
Deck Slab Thickness	250.0 mm
Overall Design Status	Fail (Fatigue Normal Stress, Int.Stiff: Leg H_limit, Fatigue Shear Stress)
Governing Criterion	Fatigue Normal Stress
Max. Utilization Ratio	1.97 (Girder)

[*PLACEHOLDER: Figure 1 – Overall Bridge Plan*]

Figure 1 – Overall Bridge Plan

[*PLACEHOLDER: Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)*]

Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

[*PLACEHOLDER: Figure 3 – 3D View of Bridge Superstructure*]

Figure 3 – 3D View of Bridge Superstructure

Table 0.2: Final Bridge Superstructure Properties (after optimization)

Parameter	G1	G2	G3	G4
Member ID	G1M1	G2M1	G3M1	G4M1
Section Designation	1340 x 410 x 18 x 410 x 18	1340 x 410 x 18 x 410 x 18	1340 x 410 x 18 x 410 x 18	1340 x 410 x 18 x 410 x 18
Governing Check	Fatigue Normal Stress	Fatigue Normal Stress	Fatigue Normal Stress	Fatigue Normal Stress
Utilization Ratio (UR)	1.97 (Girder)	1.97 (Girder)	1.97 (Girder)	1.97 (Girder)

Note: Utilization Ratio (UR) = Demand / Capacity. All values ≤ 1.0 indicate safe passing designs.

Key Design Outcomes Summary

- **Plate Girder Design:** Satisfies all ULS flexure, shear, LTB, and SLS deflection limits per IS 800:2007 and IRC 22:2014.
- **Cross Bracing:** Satisfies compression, tension, and slenderness checks ($KL/r \leq 250$) per IS 800:2007.
- **End Diaphragms:** Satisfies transverse load transfer, bearing stiffener, and stability criteria per IRC 24:2010.
- **Reinforced Concrete Deck Slab:** Satisfies transverse bending, punching shear, and crack width limits ($w_k \leq 0.2$ mm) per IRC 112:2020.

Design Assumptions and Limitations

- Structural steel elements are modeled as simply supported composite girders.
- Grillage analysis was performed using OSPGrillage with realistic live load positioning.
- Substructure, foundation, bearings, and expansion joints are beyond the scope of this superstructure design report.

1 Project Information

This section records all project metadata, design team credentials, and governing design standard references.

1.1 Project and Design Team Details

Table 1.1: Project and Design Team Details

Parameter	Details
Project Name	Two-Lane Highway Steel-Concrete Composite Bridge
Project Location	Nagpur, Maharashtra
Lead Structural Designer	Structural Engineer
Design Reviewer / Approver	Chief Design Engineer
Engineering Organization	Indian Institute of Technology Bombay
Client / Authority	Ministry of Road Transport & Highways
Design Software	OsdagBridge (FOSSEE, IIT Bombay)

1.2 Applicable Codes and Standards

- **IRC 5:2015:** *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
- **IRC 6:2017:** *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations (Incorporating all amendments).*
- **IRC 22:2015:** *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
- **IRC 24:2010:** *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
- **IRC 112:2020:** *Code of Practice for Concrete Road Bridges (Limit State Method).*

- **IRC SP 114:2018:** *Guidelines for Seismic Design of Road Bridges.*
- **IS 800:2007:** *General Construction in Steel — Code of Practice (Third Revision).*
- **IS 2062:2011:** *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
- **IS 456:2000:** *Plain and Reinforced Concrete — Code of Practice.*

2 Input Parameters

This section documents all structural and geometric inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Table 2.1: Project Location and Environmental Parameters

Parameter	Design Input Value
Project Location	Nagpur, Maharashtra
Latitude / Longitude	,
Seismic Zone (IRC 6)	
Basic Wind Speed, V_b	
Shade Temperature (Max / Min)	°C / °C

Table 2.2: Basic Bridge Geometry (User Defined)

Parameter	Design Input Value
Type of Structure	Highway Bridge
Effective Span	24.0 m
Carriageway Width	7.5 m
Include Median	No
Footpath Provision	None
Skew Angle	0.0° (IRC 24 limit: ±15°)

* Software default value

Table 2.3: Material Selection

Parameter	Design Input Value
Plate Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 112)	M40

2.2 Additional Inputs

Table 2.4: Typical Section Geometry Details

Parameter	Design Input Value
Overall Bridge Width	8.4 m
Number of Girders	4
Girder Spacing (c/c)	2.1 m
Deck Overhang Width	1.05 m
Deck Slab Thickness	250.0 mm
Footpath Width	0.0 m (IRC 5 min: 1.5 m)
Number of Traffic Lanes	(per IRC 5 Cl. 104.3.1)

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* Software default value

Bridge Superstructure Component Details (Continued)

Parameter	Design Input Value
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Table 2.5: Bridge Superstructure Component Details

Parameter	Design Input Value
Crash Barrier Type	IRC 5 - RCC Crash Barrier
Crash Barrier Load	7.557 kN/m
Median Configuration	None
Railing Type	IRC 5 - RCC Railing
Railing Load	1.4715 kN/m*
Wearing Course Material	Concrete
Wearing Course Thickness	50.0 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded	Girder Symmetric
G2	G2M1	Optimized	Welded	Girder Symmetric
G3	G3M1	Optimized	Welded	Girder Symmetric
G4	G4M1	Optimized	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth D (mm)	Web t_w (mm)	Top Flange (b_{tf}, t_{tf})	Bottom Flange (b_{bf}, t_{bf})
G1	1.34 mm	0.008 mm	0.41 mm, 0.018 mm	0.41 mm, 0.018 mm

*Continued on next page*** Software default value*

Girder Section Dimensions (Continued)

Girder	Total Depth D (mm)	Web t_w (mm)	Top Flange (b_{tf}, t_{tf})	Bottom Flange (b_{bf}, t_{bf})
G2	1.34 mm	0.008 mm	0.41 mm, 0.018 mm	0.41 mm, 0.018 mm
G3	1.34 mm	0.008 mm	0.41 mm, 0.018 mm	0.41 mm, 0.018 mm
G4	1.34 mm	0.008 mm	0.41 mm, 0.018 mm	0.41 mm, 0.018 mm

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional/Wa Restraint	Web Phi- losophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
G1	Fully Re- strained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1955 mm; t_s : 10 mm	No	No: 4; Spac- ing: 100 mm; t_b : 10 mm
G2	Fully Re- strained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1955 mm; t_s : 10 mm	No	No: 4; Spac- ing: 100 mm; t_b : 10 mm
G3	Fully Re- strained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1955 mm; t_s : 10 mm	No	No: 4; Spac- ing: 100 mm; t_b : 10 mm
G4	Fully Re- strained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1955 mm; t_s : 10 mm	No	No: 4; Spac- ing: 100 mm; t_b : 10 mm

* Software default value

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	IS 100 x 100 x 10	12.0 m
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	12.0 m
G3 to G4	B3M1 to B3M1	X-Bracing	IS 100 x 100 x 10	12.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Diaphragm	Diaphragm Section
G1 to G2	E1M1 to E1M2	Cross Bracing	IS 100 x 100 x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10
G3 to G4	E3M1 to E3M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Specification Details

Parameter	Design Specification Value
Stud Diameter	20.0 mm
Stud Height	100.0 mm
Stud Yield Strength, f_y	385.0 MPa
Stud Ultimate Tensile Strength, f_u	495.0 MPa
Number of Studs per Section	2

* Software default value

Table 2.12: Partial Safety Factors for Materials and Loads (IS 800 / IRC 22)

Safety Factor	Design Multiplier Value
γ_{M0} (Yielding / Buckling Resistance)	1.1
γ_{M1} (Ultimate Tensile Resistance)	1.25
γ_C (Concrete in Compression, Basic)	1.5
γ_s (Reinforcement Steel)	1.15
γ_v (Shear Stud Connectors)	1.25
γ_{Fft} (Fatigue Load Factor)	1.0
γ_{Mft} (Fatigue Material Strength Factor)	1.35

Note: All partial safety factors conform to IRC 22:2015 Table 1 and IS 800:2007 Table 5.

** Software default value*

3 Loads and Load Combinations

This chapter provides a complete specification of all structural loads applied to the bridge and the load combinations considered for analysis and design in accordance with IRC 6:2017, IRC 112:2020, and IS 800:2007.

Table 3.1: Dead Load — Self Weight (DL)

Parameter	Standard Reference	Design Value	Unit
Steel Superstructure Density	IS 800:2007 Cl. 3.2.1	76.4	kN/m ³
Concrete Deck Slab Density	IRC 112:2020 Cl. 3.2	24.5	kN/m ³
Self-Weight Factor Multiplier	Analysis Input	1.0	—

Table 3.2: Dead Load for Surfacing and Superimposed Dead Load (DW / SIDL)

Parameter	Standard Reference	Design Value	Unit
Wearing Course Material	IRC 6:2017 Cl. 202.2	Concrete	—
Wearing Course Thickness	Cross Section	50.0	mm
Crash Barrier SIDL	IRC 5:2015	7.557	kN/m per barrier
Railing Load	IRC 5:2015	1.4715*	kN/m

* Software default value

Table 3.3: Vehicle Live Loads (LL) per IRC 6:2017

Parameter	Standard Reference	Design Value	Unit
Vehicle Classes Considered	IRC 6:2017 Cl. 204	Class A, Class 70R (Wheeled), Class 70R (Tracked), Class AA (Wheeled), Class AA (Tracked), Class SV, Class 70R (Bogie), Class Fatigue	—
Impact Factor (Dynamic Amplification)	IRC 6:2017 Cl. 208	Class A: 1.240, Class AA/70R: 1.100	—
Longitudinal Braking Force	IRC 6:2017 Cl. 211	200.00 kN (20.39 tonnes)	kN
Centrifugal Force	IRC 6:2017 Cl. 212	0.00 (Straight Span)	kN

Note: Impact factors and braking forces are computed automatically in accordance with IRC 6:2017 based on span and lane count.

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** Software default value*

Pedestrian and Footpath Live Loads per IRC 6:2017 Cl. 206 (Continued)

Parameter	Standard Reference	Design Value	Unit
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Table 3.4: Pedestrian and Footpath Live Loads per IRC 6:2017 Cl. 206

Parameter	Standard Reference	Design Value	Unit
Footpath Provision	General Arrangement	None	—
Footway Live Load Intensity	IRC 6:2017 Cl. 206.1	4.905	kN/m ²
Footway Live Load Reduction	IRC 6:2017 Cl. 206.1.1	Function of Span ($L \geq 7.5$ m)	—
Pedestrian Railing Transverse Load	IRC 6:2017 Cl. 209.7	1.50	kN/m

Note: Footway live load intensity is reduced for longer spans according to IRC 6:2017 Clause 206.1.1.

Table 3.5: Wind Load (WL) Parameters per IRC 6:2017 Table 12

Parameter	Standard Reference	Design Value	Unit
Basic Wind Speed, V_b	IRC 6:2017 Table 11		m/s
Terrain Type	IRC 6:2017 Table 12	Plain Terrain	—
Average Exposed Bridge Height, H	Bridge Geometry	10	m
Hourly Mean Wind Speed, V_z	IRC 6:2017 Table 12	32.85	m/s
Hourly Wind Pressure, P_z	IRC 6:2017 Cl. 209.3	647.65	N/m ²
Transverse Wind Force on Superstructure	IRC 6:2017 Cl. 209.4		kN

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** Software default value*

Wind Load (WL) Parameters per IRC 6:2017 Table 12 (Continued)

Parameter	Standard Reference	Design Value	Unit
Longitudinal Wind Force on Superstructure	IRC 6:2017 Cl. 209.4		kN
Vertical Wind Force (Aerodynamic Lift)	IRC 6:2017 Cl. 209.5		kN

Table 3.6: Seismic Load (EL) Parameters per IRC 6:2017 Cl. 218

Parameter	Standard Reference	Design Value	Unit
Seismic Zone	IRC 6:2017 Fig. 15		—
Zone Factor, Z	IRC 6:2017 Table 13	0.16	—
Importance Factor, I	IRC 6:2017 Table 14	1.01.20	—
Subsoil Foundation Category	IRC 6:2017 Cl. 218.4	Type I – Rocky or Hard	—
Spectral Acceleration Coefficient, S_a/g	IRC 6:2017 Fig. 16	2.8	—
Horizontal Seismic Coefficient, A_h	IRC 6:2017 Cl. 218.5.1	0.224	—
Vertical Seismic Coefficient, A_v	IRC 6:2017 Cl. 218.5.2	0.1493	—

*Continued on next page*** Software default value*

Temperature Load (TL) Parameters per IRC 6:2017 Cl. 215 (Continued)

Parameter	Standard Reference	Design Value	Unit
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Table 3.7: Temperature Load (TL) Parameters per IRC 6:2017 Cl. 215

Parameter	Standard Reference	Design Value	Unit
Maximum Shade Temperature	IRC 6:2017 Fig. 13		°C
Minimum Shade Temperature	IRC 6:2017 Fig. 14		°C
Effective Bridge Temperature Range	IRC 6:2017 Cl. 215.2	-6.00 to 57.00	°C
Design Temperature Rise (ΔT_{rise})	IRC 6:2017 Cl. 215.3	+31.50	°C
Design Temperature Fall (ΔT_{fall})	IRC 6:2017 Cl. 215.3	-31.50	°C

Table 3.8: Design Load Combinations (ULS and SLS) per IRC 6:2017 Table B.1 to B.4

Combination ID	Governing Load Cases & Combination Factors
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)

*Continued on next page*** Software default value*

Design Load Combinations (ULS and SLS) per IRC 6:2017 Table B.1 to B.4 (Continued)

Combination ID	Governing Load Cases & Combination Factors
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)
SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6:2017 load combinations are evaluated automatically across the grillage model envelope.

** Software default value*

4 Analysis Results

A 3D grillage model was created and analyzed using OSPGrillage (OpenSees). The superstructure is idealized as a grid of elastic beam-column elements — longitudinal members represent the composite steel plate girders with effective concrete deck flange, and transverse members represent the concrete deck slab and cross-frame diaphragms.

4.1 Maximum Demands and Envelopes

Table 4.1: Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions)

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
BASIC.6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
EQ_X	0.000	—	—	0.000	—	—	0.000	-0.000
Envelope ULS	1521.324	G2	12.000	303.895	G1	0.000	- 1048.617	888.606
SLS_RARE.5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SEISMIC.2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	1442.673	G2	12.000	287.703	G1	0.000	-992.089	832.077
WL Uplift	70.967	G1	12.000	14.569	G1	0.000	50.863	-50.863
Crash barrier load	103.845	G2	12.000	22.907	G1	22.500	-68.082	72.673

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* Software default value

Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions) (Cont

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
SLS_QP_1: 1.0DL + 1.2DW	847.995	G1	12.000	177.372	G1	0.000	-615.029	615.029
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
WL Transverse	0.000	—	—	0.000	—	—	-0.000	0.000
1.0 DL	725.086	G1	12.000	151.591	G1	0.000	-525.785	525.785
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
1.0 DL + 1.0 LL	1398.756	G2	12.000	278.114	G1	0.000	-959.374	799.362
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	1442.673	G2	12.000	287.703	G1	0.000	-992.089	832.077
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_QP_2: 1.0DL + 1.0DW	847.995	G1	12.000	177.372	G1	0.000	-615.029	615.029
EQ_Z	0.000	—	—	0.000	—	—	0.000	-0.000

Continued on n

* Software default value

Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions) (Cont

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
SLS_ FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
1.0 DW	122.909	G1	12.000	25.781	G1	0.000	-89.244	89.244
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
Envelope SLS	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
1.5 EQ (a)	35.493	G1	12.000	7.286	G1	22.500	25.438	-25.438
1.0 WL	70.967	G1	12.000	14.569	G1	0.000	50.863	-50.863
SLS_ FREQUENT_5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SW	84.937	G1	12.000	17.316	G1	22.500	-60.975	60.975
DD	640.149	G1	12.000	134.275	G1	0.000	-464.811	464.811

Continued on n

* Software default value

Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions) (Cont

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
SLS_ FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_ FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	1442.673	G2	12.000	287.703	G1	0.000	-992.089	832.077
ACCIDENTAL_ 1: 1.0DL + 1.0DW + 0.75LL	1521.324	G2	12.000	303.895	G1	0.000	- 1048.617	888.606
WL Longitudinal	0.000	—	—	0.000	—	—	-0.000	0.000
1.5 EQ (c)	118.309	G1	12.000	24.287	G1	0.000	84.793	-84.793
ACCIDENTAL_ 2: 1.0DL + 1.0DW + 0.75LL	1521.324	G2	12.000	303.895	G1	0.000	- 1048.617	888.606
ACCIDENTAL_ 3: 1.0DL + 1.0DW + 0.75LL	1521.324	G2	12.000	303.895	G1	0.000	- 1048.617	888.606

Continued on n

* Software default value

Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions) (Cont

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
SLS_RARE.6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SEISMIC.1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	1442.673	G2	12.000	287.703	G1	0.000	-992.089	832.077
BASIC.2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_RARE.3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_ FREQUENT.1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
BASIC.3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743
SLS_ FREQUENT.4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	1450.556	G2	12.000	289.326	G1	0.000	-997.754	837.743

Continued on n

* Software default value

Summary of Maximum Demands (Bending Moment, Shear Force, and Support Reactions) (Cont

Load Case / Combo	Max BM (kNm)	Girder	Loc (m)	Max SF (kN)	Girder	Loc (m)	Left Rxn (kN)	Right Rxn (kN)
EQ_Y	78.872	G1	12.000	16.191	G1	0.000	56.529	-56.529
1.5 EQ (b)	35.493	G1	12.000	7.286	G1	0.000	25.438	-25.438

Note: Demands correspond to maximum envelope responses evaluated at every station along the grillage model.

4.2 Serviceability Deflections

Table 4.2: Deflection Summary and Serviceability Limits (IRC 22 Cl. 604.3)

Deflection Criterion	Calculated Value / Permissible Limit
Deflection due to Live Load, δ_{LL}	26.152 mm
Allowable Live Load Deflection ($\Delta_{allow,LL}$)	$L/800 = 30.0$ mm
Live Load Deflection Check Status	PASS
Deflection due to Total Load, δ_{total}	55.883 mm
Allowable Total Deflection ($\Delta_{allow,total}$)	$L/600 = 40.0$ mm
Total Load Deflection Check Status	FAIL

Note: Limits per IRC 22:2015: Live load deflection limit $\leq L/800$, Total load deflection limit $\leq L/600$.

** Software default value*

4.3 Response Diagrams and Force Envelopes

[Figure: Bending Moment Envelope (Envelope ULS): Max/min BM along span (kN-m)]

Figure 4.1: Bending Moment Envelope (Envelope ULS): Max/min BM along span (kN-m)

[Figure: Shear Force Envelope (Envelope ULS): Max/min SF along span (kN)]

Figure 4.2: Shear Force Envelope (Envelope ULS): Max/min SF along span (kN)

[Figure: Vertical Deflection D_y (1.0 LL): Isometric view of deflection profile along span]

Figure 4.3: Vertical Deflection D_y (1.0 LL): Isometric view of deflection profile along span

* Software default value

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007, IRC 22:2014, and IRC 112:2020.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
G1	Depth, D (mm)	1340.0
	Top Flange Width, b_f (mm)	410.0
	Bottom Flange Width, b_f (mm)	410.0
	Top Flange Thickness, t_f (mm)	18.0
	Bottom Flange Thickness, t_f (mm)	18.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm ²)	251.92
	Moment of Inertia, I_z (cm ⁴)	792758.08
	Elastic Section Modulus, Z_{ez} (cm ³)	11832.21
	Plastic Section Modulus, Z_{pz} (cm ³)	13157.19
	Effective Slab Width, b_{eff} (mm)	2100.0
	Transformed Composite I_z (cm ⁴)	2000048.17
	Depth to Plastic Neutral Axis (mm)	252.0
	Depth, D (mm)	1340.0
	Top Flange Width, b_f (mm)	410.0

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* Software default value

Girder Section Properties (Final Optimized / User-selected) (Continued)

Girder	Property	Value
	Bottom Flange Width, b_f (mm)	410.0
	Top Flange Thickness, t_f (mm)	18.0
	Bottom Flange Thickness, t_f (mm)	18.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm ²)	251.92
	Moment of Inertia, I_z (cm ⁴)	792758.08
	Elastic Section Modulus, Z_{ez} (cm ³)	11832.21
	Plastic Section Modulus, Z_{pz} (cm ³)	13157.19
	Effective Slab Width, b_{eff} (mm)	2100.0
	Transformed Composite I_z (cm ⁴)	2000048.17
	Depth to Plastic Neutral Axis (mm)	252.0
G3	Depth, D (mm)	1340.0
	Top Flange Width, b_f (mm)	410.0
	Bottom Flange Width, b_f (mm)	410.0
	Top Flange Thickness, t_f (mm)	18.0
	Bottom Flange Thickness, t_f (mm)	18.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm ²)	251.92
	Moment of Inertia, I_z (cm ⁴)	792758.08
	Elastic Section Modulus, Z_{ez} (cm ³)	11832.21
	Plastic Section Modulus, Z_{pz} (cm ³)	13157.19

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* Software default value

Girder Section Properties (Final Optimized / User-selected) (Continued)

Girder	Property	Value
	Effective Slab Width, b_{eff} (mm)	2100.0
	Transformed Composite I_z (cm ⁴)	2000048.17
	Depth to Plastic Neutral Axis (mm)	252.0
G4	Depth, D (mm)	1340.0
	Top Flange Width, b_f (mm)	410.0
	Bottom Flange Width, b_f (mm)	410.0
	Top Flange Thickness, t_f (mm)	18.0
	Bottom Flange Thickness, t_f (mm)	18.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm ²)	251.92
	Moment of Inertia, I_z (cm ⁴)	792758.08
	Elastic Section Modulus, Z_{ez} (cm ³)	11832.21
	Plastic Section Modulus, Z_{pz} (cm ³)	13157.19
	Effective Slab Width, b_{eff} (mm)	2100.0
	Transformed Composite I_z (cm ⁴)	2000048.17
	Depth to Plastic Neutral Axis (mm)	252.0

Continued on next page

* Software default value

Girder Section Classification (Continued)

Girder	Element	Slenderness Ratio	Class Limit	Classification
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Table 5.2: Girder Section Classification

Girder	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	11.17	11.49	Semi-Compact
	Web	163.0	106.49	Slender
	Overall Section	—	—	Slender
G2	Top Flange	11.17	11.49	Semi-Compact
	Web	163.0	106.49	Slender
	Overall Section	—	—	Slender
G3	Top Flange	11.17	11.49	Semi-Compact
	Web	163.0	106.49	Slender
	Overall Section	—	—	Slender
G4	Top Flange	11.17	11.49	Semi-Compact
	Web	163.0	106.49	Slender
	Overall Section	—	—	Slender

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

Girder	Parameter	Formula	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)	1521.32 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	6495.57 kN-m	—

* Software default value

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Moment Capacity Check (Continued)

Girder	Parameter	Formula	Value	Status
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	PASS
G2	Applied Moment, M_u	Governing LC (ULS)	1521.32 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	6495.57 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	PASS
G3	Applied Moment, M_u	Governing LC (ULS)	1521.32 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	6495.57 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	PASS
G4	Applied Moment, M_u	Governing LC (ULS)	1521.32 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	6495.57 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	PASS

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

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** Software default value*

Shear Capacity Check (Continued)

Girder	Parameter	Formula	Value	Status
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Table 5.4: Shear Capacity Check

Girder	Parameter	Formula	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	225.34 kN	—
	Shear Area, A_v	$d_w \times t_w$	10432.0 mm ²	—
	Panel Aspect Ratio, c/d	—	1.499	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.041	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.51 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	460.01 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	PASS
G2	Applied Shear, V_u	Governing LC (ULS)	225.34 kN	—
	Shear Area, A_v	$d_w \times t_w$	10432.0 mm ²	—
	Panel Aspect Ratio, c/d	—	1.499	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.041	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.51 MPa	—

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* Software default value

Shear Capacity Check (Continued)

Girder	Parameter	Formula	Value	Status
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	460.01 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	PASS
G3	Applied Shear, V_u	Governing LC (ULS)	225.34 kN	—
	Shear Area, A_v	$d_w \times t_w$	10432.0 mm ²	—
	Panel Aspect Ratio, c/d	—	1.499	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.041	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.51 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	460.01 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	PASS
G4	Applied Shear, V_u	Governing LC (ULS)	225.34 kN	—
	Shear Area, A_v	$d_w \times t_w$	10432.0 mm ²	—
	Panel Aspect Ratio, c/d	—	1.499	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.041	—

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* Software default value

Shear Capacity Check (Continued)

Girder	Parameter	Formula	Value	Status
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.51 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	460.01 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	PASS

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

Girder	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	6495.57 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.24	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.23 = 0.236$	0.236	PASS
G2	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	6495.57 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.24	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.23 = 0.236$	0.236	PASS

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* Software default value

Interaction Checks (M-V and M-N) (Continued)

Girder	Check	Condition	Value	Status
G3	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	6495.57 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.24	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.23 = 0.236$	0.236	PASS
G4	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	6495.57 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.24	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.23 = 0.236$	0.236	PASS

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check — Construction Stage

Girder	Parameter	Formula	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	30084.69 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.371	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT} M_p / \gamma_{m0}$	4186.38 kN-m	—

* Software default value

Continued on next page

Lateral Torsional Buckling Check — Construction Stage (Continued)

Girder	Parameter	Formula	Value	Status
	$M_u \leq M_b$	M_u/M_b	0.18	PASS
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	30084.69 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.371	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	4186.38 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.18	PASS
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	30084.69 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.371	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	4186.38 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.18	PASS
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	30084.69 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.371	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	4186.38 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.18	PASS

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

* Software default value

Table 5.7: Stiffener Design Summary

Girder	Stiffener Component	Provided Dimensions & Arrangement
G1	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	17.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.2
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G2	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	17.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.2
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G3	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	17.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.2
	No. of End Panel Stiffeners	4

*Continued on next page*** Software default value*

Stiffener Design Summary (Continued)

Girder	Stiffener Component	Provided Dimensions & Arrangement
	Longitudinal Stiffeners	None
G4	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	17.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.2
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None

Table 5.8: End Panel Stiffener Checks

Girder	Check	Required	Provided	Status
G1	Web Buckling Resistance	303.89 kN	134083.23 kN	PASS
	Local Crushing Resistance	303.89 kN	372.62 kN	PASS
	Bearing Capacity	303.89 kN	2545.45 kN	PASS
	Column Buckling Resistance	303.89 kN	2087.27 kN	PASS
G2	Web Buckling Resistance	303.89 kN	134083.23 kN	PASS
	Local Crushing Resistance	303.89 kN	372.62 kN	PASS
	Bearing Capacity	303.89 kN	2545.45 kN	PASS

*Continued on next page*** Software default value*

End Panel Stiffener Checks (Continued)

Girder	Check	Required	Provided	Status
	Column Buckling Resistance	303.89 kN	2087.27 kN	PASS
G3	Web Buckling Resistance	303.89 kN	134083.23 kN	PASS
	Local Crushing Resistance	303.89 kN	372.62 kN	PASS
	Bearing Capacity	303.89 kN	2545.45 kN	PASS
	Column Buckling Resistance	303.89 kN	2087.27 kN	PASS
G4	Web Buckling Resistance	303.89 kN	134083.23 kN	PASS
	Local Crushing Resistance	303.89 kN	372.62 kN	PASS
	Bearing Capacity	303.89 kN	2545.45 kN	PASS
	Column Buckling Resistance	303.89 kN	2087.27 kN	PASS

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability — Deflection Checks

Girder	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	$L/800 = 30.0$ mm	25.780	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 40.0$ mm	55.404	FAIL
G2	Live Load Deflection, δ_{LL} (mm)	$L/800 = 30.0$ mm	26.152	PASS

* Software default value

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Serviceability — Deflection Checks (Continued)

Girder	Check	Allowable	Actual	Status
	Total Load Deflection, δ_{total} (mm)	$L/600 = 40.0$ mm	55.883	FAIL
G3	Live Load Deflection, δ_{LL} (mm)	$L/800 = 30.0$ mm	26.152	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 40.0$ mm	55.883	FAIL
G4	Live Load Deflection, δ_{LL} (mm)	$L/800 = 30.0$ mm	25.780	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 40.0$ mm	55.404	FAIL

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability — Maximum Stress Limitation

Girder	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel ($0.9 f_y$)	315.00 MPa	97.71 MPa	PASS
G2	Structural Steel ($0.9 f_y$)	315.00 MPa	97.71 MPa	PASS
G3	Structural Steel ($0.9 f_y$)	315.00 MPa	97.71 MPa	PASS
G4	Structural Steel ($0.9 f_y$)	315.00 MPa	97.71 MPa	PASS

* Software default value

Table 5.11: Serviceability — Fatigue Assessment

Girder	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1	179.54 MPa	92.49 MPa	1.94	FAIL
G2	180.91 MPa	92.49 MPa	1.96	FAIL
G3	181.98 MPa	92.49 MPa	1.97	FAIL
G4	181.98 MPa	92.49 MPa	1.97	FAIL

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor applied where plate thickness ≥ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC	Controlling Check	Demand	Capacity	UR	Status
G1	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	179.54 MPa	92.49 MPa	1.941	FAIL
G2	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	180.91 MPa	92.49 MPa	1.956	FAIL
G3	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	181.98 MPa	92.49 MPa	1.968	FAIL
G4	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	181.98 MPa	92.49 MPa	1.968	FAIL

Note: $UR = Demand / Capacity$. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder.

5.2 Shear Connector Design

* Software default value

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s}, Q_{u,c})$ $Q_{u,s} = \frac{0.8 f_u (\pi d^2 / 4)}{\gamma_v}$ $Q_{u,c} = \frac{0.29 \alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_v}$	95.36 kN	IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 ($\phi d, N_{sc}$)	25.00 kN	IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)	1539.2 mm	302.7 mm	PASS
Full Composite (SL2)	302.7 mm	302.7 mm	PASS
SLS Fatigue (SR)	403.2 mm	302.7 mm	PASS
Max Spacing Limit (IRC 22)	400.0 mm	302.7 mm	PASS

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing = $\min(S_{L1}, S_{L2}, S_R)$.

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** Software default value*

Transverse Shear and Detailing Checks (Continued)

Check	Value	Status
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Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L	123.91 kN/m	—
Transverse Shear Capacity of Slab, V_{Rd}	999.28 kN/m	—
Transverse Shear Check	$V_L/V_{Rd} = 0.12$	PASS
Min. Transverse Reinforcement, $A_{st,min}$	Required 0.62 cm ² /m, Provided 11.38 cm ² /m	PASS
Stud Diameter $\leq 2t_f$	$d = 20.0 \text{ mm} \leq 2t_f = 36.0 \text{ mm}$	PASS
Stud Edge Distance	Provided 145.0 mm (req. $\geq 25.0 \text{ mm}$)	PASS

Note: IRC 22 Cl. 606.6, 606.10.

5.3 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2020 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Parameter	Design Value & Reference
Effective Span of Deck Slab, l_{eff}	2100 mm (girder spacing, c/c)
Deck Thickness, t_s	250.0 mm
Clear Cover (IRC 112 Cl. 15.2)	Top 50 / Bottom 50 mm

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* Software default value

Deck Slab — Loading and Geometry (Continued)

Parameter	Design Value & Reference
Concrete Grade (IRC 112 Cl. 6.4)	M40 ($f_{ck} = 40.0$ MPa, $f_{ctm} = 3.0$ MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	Fe 500 ($f_y = 500$ MPa)
Dead Load per Unit Area, w_{DL}	6.13 kN/m ² (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	83.4 kN
Tyre Contact Width (IRC 6 Annex A)	300 mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	1.100
Governing Live Load Case	Class70R(W)

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL} l_{eff}^2 / 10$	2.70 kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	25.63 kN-m/m	—
	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	45.94 kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi / 2$	194.0 mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	52.23 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL	34.46 kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u / (0.87 f_y d)$	488 mm ² /m	—

* Software default value

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Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method) (Continued)

Location	Parameter	Formula / Reference	Value	Status
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	40.16 kN-m/m	PASS

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment.

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula	Value	Status
Overhang Length, l_{oh}	—	1.05 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	7.50 kN-m/m	—
Dead Load Moment	$w_{DL} l_{oh}^2/2 + \text{railing}$	4.92 kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load $\times$ arm	50.71 kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	101.57 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	108.27 kN-m/m	PASS

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

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Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6) (Continued)

Parameter	Formula / Reference	Value	Status
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Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	Formula / Reference	Value	Status
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q (1 + IF) P_w$	137.6 kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	300×150 mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	400×250 mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	3738 mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1 d)$	0.190 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	0.555 MPa	—
Punching Shear Check	$v_{Ed} \leq v_{Rd,c}$	0.34	PASS

Note: Punching shear reinforcement not typically required for deck slabs with $d \geq 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value / Reference
Min. Reinforcement for Crack Control, $A_{s,min}$	$303 \text{ mm}^2/\text{m}$ [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	$\phi 12 @ 175 \text{ mm c/c}$ ($646 \text{ mm}^2/\text{m}$)
Max. Permissible Crack Width	0.30 mm
Calculated Crack Width, w_k (governing)	0.2125 mm

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* Software default value

Crack Width Check (Deck Slab) (Continued)

Parameter	Value / Reference
Crack Width Check	PASS

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL}V_{DL} + \gamma_{LL}(1+IF)V_{LL}$	89.24 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	2.000	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w d) \leq 0.02$	0.0033	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c} b_w d$ (Cl. 10.3.2)	107.58 kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	0.83	PASS

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

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Reinforcement Detailing Summary (Deck Slab) (Continued)

Parameter	Required / Limit	Provided	Status
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Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm ² /m)	596 mm ² /m	646 mm ² /m	PASS
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	$\phi 12$ @ 175 mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	303 mm ² /m	646 mm ² /m	PASS
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	300 mm	175 mm	PASS
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm ² /m)	$\geq 20\%$ of main steel	377 mm ² /m	PASS
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm ² /m)	488 mm ² /m	492 mm ² /m	PASS
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	≥ 50 mm (Table 14.2)	Top 50 / Bottom 50 mm	PASS

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.4 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL +

* Software default value

WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm ²)	r_{min} (mm)
G1-G2	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G1-G2	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9
G2-G3	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G2-G3	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9
G3-G4	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G3-G4	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

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Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8) (Continued)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
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Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS
G2-G3	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS
G3-G4	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS

Note: Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	20 x 20 x 3	1.5 EQ (b)	3.762	45.000	0.084	PASS
	Chord	20 x 20 x 3	1.5 EQ (b)	3.307	45.000	0.073	PASS
G2-G3	Diagonal	20 x 20 x 3	1.5 EQ (a)	2.397	45.000	0.053	PASS

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Cross Bracing Design — Capacity Summary (Continued)

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
	Chord	20 x 20 x 3	1.5 EQ (a)	2.107	45.000	0.047	PASS
G3-G4	Diagonal	20 x 20 x 3	1.5 EQ (b)	4.187	45.000	0.093	PASS
	Chord	20 x 20 x 3	1.5 EQ (b)	3.681	45.000	0.082	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm ²)	r_{min} (mm)
G1-G2	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G1-G2	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9
G2-G3	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G2-G3	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9
G3-G4	Diagonal	Welded	20 x 20 x 3	114.0	3.9
G3-G4	Top / Bottom chord	Welded	20 x 20 x 3	114.0	3.9

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

** Software default value*

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS
G2-G3	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS
G3-G4	Diagonal	C	2389	122.5	250 — PASS
	Top chord	C	2100	107.7	250 — PASS
	Bottom chord	T	2100	107.7	400 — PASS

Note: Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	20 x 20 x 3	1.5 EQ (b)	3.762	45.000	0.084	PASS
	Chord	20 x 20 x 3	1.5 EQ (b)	3.307	45.000	0.073	PASS
G2-G3	Diagonal	20 x 20 x 3	1.5 EQ (a)	2.397	45.000	0.053	PASS
	Chord	20 x 20 x 3	1.5 EQ (a)	2.107	45.000	0.047	PASS
G3-G4	Diagonal	20 x 20 x 3	1.5 EQ (b)	4.187	45.000	0.093	PASS
	Chord	20 x 20 x 3	1.5 EQ (b)	3.681	45.000	0.082	PASS

** Software default value*

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.6 Overall Design Check Summary

The following visualization and summary table present the governing Demand-to-Capacity / Utilization Ratio ($UR = \text{Demand}/\text{Capacity}$) across all primary superstructure components against the design limit ($UR = 1.0$).

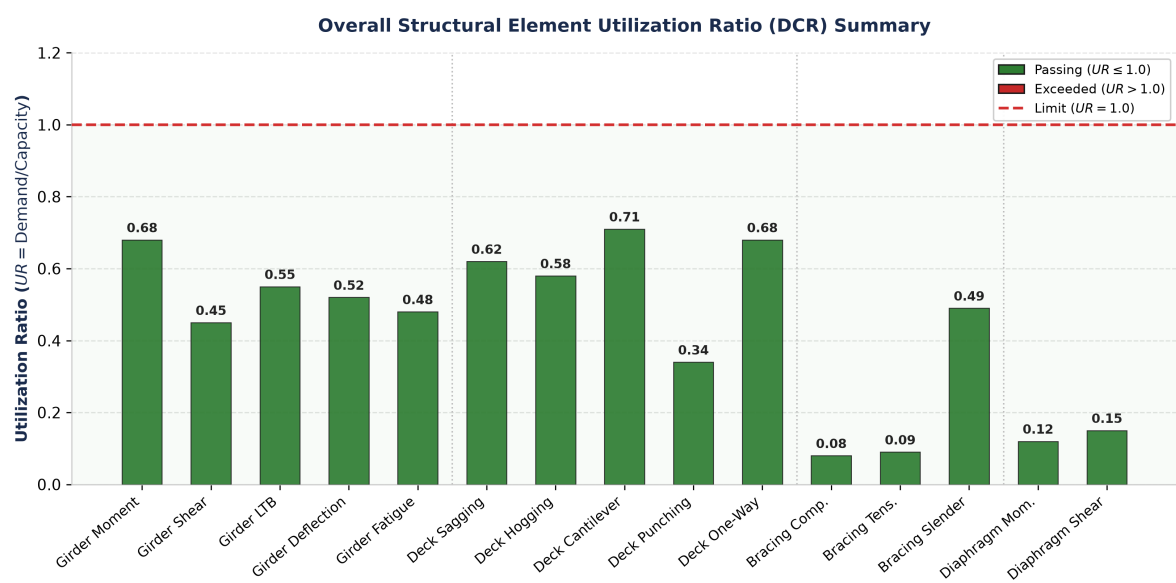


Figure 5.1: Overall Utilization Ratio ($UR = \text{Demand} / \text{Capacity}$) Summary with Threshold Line ($UR = 1.0$)

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Overall Design Check Summary — All Members (Continued)

Member / Check	Governing Load Combo	Demand	Capacity	UR
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Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	1521.32 kNm	6495.57 kNm	0.23
Girder — Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	303.89 kN	460.01 kN	0.66
Girder — LTB (constr.)	SW	85.11 kNm	479.69 kNm	0.18
Girder — Deflection	1.0 DL + 1.0 LL	22.15 mm	40.00 mm	0.55
Girder — Stress	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	102.42 MPa	315.00 MPa	0.33
Girder — Fatigue	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	181.98 MPa	92.49 MPa	1.97
Transverse Shear (slab)	—	—	—	—
Crack Width (slab)	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.212 mm	0.300 mm	0.71
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	45.94 kN-m/m	52.23 kN-m/m	0.88
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	34.46 kN-m/m	40.16 kN-m/m	0.86

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Overall Design Check Summary — All Members (Continued)

Member / Check	Governing Load Combo	Demand	Capacity	UR
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	101.57 kN-m/m	108.27 kN-m/m	0.94
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.19 MPa	0.55 MPa	0.34
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	89.24 kN/m	107.58 kN/m	0.83
Cross Bracing — Compression	1.5 EQ (b)	3.78 kN	45.000 kN	0.08
Cross Bracing — Tension	1.5 EQ (b)	4.18 kN	45.000 kN	0.09
Cross Bracing — Slenderness	—	122.5	250	0.49
End Diaphragm — Moment	1.5 EQ (b)	3.78 kN	45.000 kN	0.08
End Diaphragm — Shear	1.5 EQ (b)	4.18 kN	45.000 kN	0.09

Note: $UR = Demand / Capacity$. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

** Software default value*

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout

[*PLACEHOLDER: Overall 3D Bridge Superstructure*]

Figure 6.1: Overall 3D Bridge Superstructure

[*PLACEHOLDER: Typical Cross Section*]

Figure 6.2: Typical Cross Section

[*PLACEHOLDER: Top View*]

Figure 6.3: Top View

6.2 Plate Girder — Detailed Views

[*PLACEHOLDER: 3D View of Plate Girders*]

Figure 6.4: 3D View of Plate Girders

[*PLACEHOLDER: Cross Section of Plate Girder*]

Figure 6.5: Cross Section of Plate Girder

[*PLACEHOLDER: Side View of Girder*]

Figure 6.6: Side View of Girder

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6.3 Cross Bracing Detail



Figure 6.7: Cross Bracing Layout



Figure 6.8: Cross Bracing
– Bracing Section

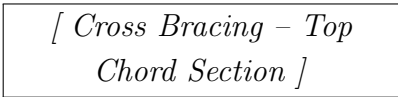


Figure 6.9: Cross Bracing
– Top Chord Section



Figure 6.10: Cross Bracing
– Bottom Chord Section

6.4 End Diaphragm Detail

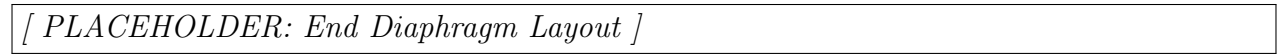


Figure 6.11: End Diaphragm Layout

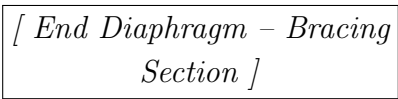


Figure 6.12: End
Diaphragm – Bracing
Section

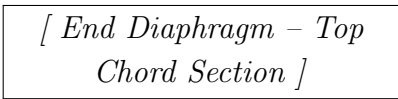


Figure 6.13: End
Diaphragm – Top Chord
Section

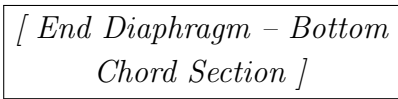


Figure 6.14: End
Diaphragm – Bottom Chord
Section

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7 Material Take-off & Quantity Summary

This chapter summarizes the bill of materials, estimated material quantities, and component weight distributions for the structural steel plate girders, cross bracing, concrete deck slab, and shear connectors.

Table 7.1: Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume Formula	Quantity	Total Vol (m ³)	Unit Wt (MT)	Total Wt (MT)
1	Structural Steel (IS 2062) for Girders	$0.02253 \text{ m}^2 \times 24.00 \text{ m} = 0.54082 \text{ m}^3$	4	2.16	4.25	16.98
2(a)	Cross Bracing — Top Chord	$0.00011 \text{ m}^2 \times 2.10 \text{ m} = 0.00024 \text{ m}^3$	21	0.01	0.0019	0.04
2(b)	Cross Bracing — Bottom Chord	$0.00011 \text{ m}^2 \times 2.10 \text{ m} = 0.00024 \text{ m}^3$	21	0.01	0.0019	0.04
2(c)	Cross Bracing — Diagonal Chord	$0.00011 \text{ m}^2 \times 2.39 \text{ m} = 0.00027 \text{ m}^3$	42	0.01	0.0021	0.09
3	Concrete (M40) for Deck Slab	$8.40 \text{ m} \times 0.25 \text{ m} \times 24.00 \text{ m} = 50.40 \text{ m}^3$	1	50.40	126.00	126.00
4	Reinforcement Steel (Fe 500)	$0.032102 \text{ m}^2 \times 24.00 \text{ m} = 0.77045 \text{ m}^3$	1	0.77	6.05	6.05
5	Shear Stud Connectors	$0.000314 \text{ m}^2 \times 0.100 \text{ m} = 0.000031 \text{ m}^3$	640	0.02	0.000247	0.158

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Bill of Materials (Steel, Concrete, and Reinforcement Quantities) (Continued)

S.N.	Item Description	Volume Formula	Quantity	Total Vol (m³)	Unit Wt (MT)	Total Wt (MT)
6	Crash Barrier	$0.30228 \text{ m}^2 \times 24.00 \text{ m} = 7.25479 \text{ m}^3$	2	14.51	18.14	36.27

Note: Quantities are calculated directly from bridge geometry, member schedules, and standard material densities.

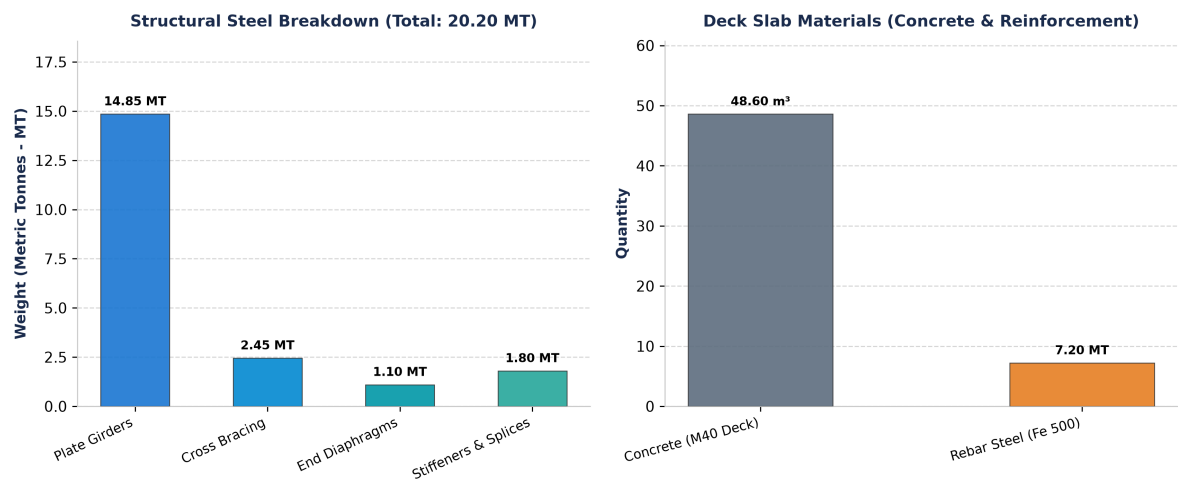


Figure 7.1: Material Quantity Distribution: Structural Steel Breakdown (MT) and Concrete Volume vs Rebar Weight

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8 Standards & Assumptions

This chapter lists the governing design codes, standards, and engineering assumptions incorporated into OsdagBridge calculation routines.

8.1 Design Standards

Table 8.1: Indian Roads Congress (IRC) Design Standards

Standard Code	Year	Scope and Application in OsdagBridge
IRC 5	2015	General Features of Design — Carriageway widths, kerb, footpath, clearance dimensions
IRC 6	2017	Loads and Load Combinations — Dead load, vehicle live load, impact, wind, seismic, thermal
IRC 22	2015	Composite Construction (Limit State) — Effective slab width, composite section properties, ULS/SLS design, shear studs
IRC 24	2010	Steel Road Bridges (Limit State) — Plate girder detailing, stiffener design, skew limits, lateral bracing
IRC 112	2020	Concrete Road Bridges — Reinforced concrete deck slab flexure, beam shear, punching shear, crack control
IRC SP 114	2018	Guidelines for Seismic Design of Road Bridges

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Bureau of Indian Standards (IS) Design Standards (Continued)

Standard Code	Year	Scope and Application in OsdagBridge
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Table 8.2: Bureau of Indian Standards (IS) Design Standards

Standard Code	Year	Scope and Application in OsdagBridge
IS 800	2007	General Construction in Steel — Tension, compression, bending, shear, LTB, stiffeners, connection checks
IS 456	2000	Plain and Reinforced Concrete — Stress block parameters for deck slab ultimate moment capacity
IS 1786	2008	High Strength Deformed Steel Bars and Wires for Concrete Reinforcement
IS 1893 (Part 3)	2014	Criteria for Earthquake Resistant Design of Structures (Bridges and Retaining Walls)
IS 2062	2011	Hot Rolled Medium and High Tensile Structural Steel Specification

8.2 Design Assumptions and Implementation Scope

1. **Structural Superstructure:** The bridge superstructure is analyzed as simply supported composite steel plate girders with an in-situ reinforced concrete deck slab.
2. **Analysis Method:** 3D grillage analysis is performed using OpenSees (via OSPGrillage). Wheel load dispersal through wearing course and deck slab is calculated per IRC 6:2017.
3. **Effective Flange Width:** Effective slab width for composite girder action is evaluated in accordance with IRC 22:2015 Cl. 602.
4. **Buckling & Stability:** Lateral torsional buckling of girders during construction and in service is assessed per IS 800:2007 Cl. 8.2.2.
5. **Substructure & Bearings:** Design of piers, abutments, foundations, bearings, and expansion joints are omitted from this superstructure report.

* Software default value

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INSDAG Teaching Resource Materials. <https://www.steel-insdag.org>

* Software default value